

# Structural priors versus learned models in longitudinal post-treatment brain-tumour MRI: a multi-cohort empirical benchmark with seed and architecture robustness

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## Highlights

- Multi-cohort empirical benchmark on 522 paired post-treatment brain-tumour MRIs
- Five seeds  $\times$  four architectures (incl. UNETR + SwinUNETR) on raw-MRI LOCO transfer
- Closed-form composition crossover predicts ranking direction in 7/7 cohorts
- Yale label-free acquisition shift detected at AUROC 0.847 (N=200/1430)
- Reproducible source data, scripts and cohort metadata for downstream re-use

(5 bullets; longest 76 chars; within 85-char MedIA limit.)

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## Graphical Abstract

A three-panel composite at 300 DPI ( $531 \times 1328$  pixels): (a) per-cohort held-out Brier across heat-kernel prior, lightweight 3D U-Net, residual 3D ResUNet (with calibration + TTA), UNETR transformer and SwinUNETR transformer; (b) raw+mask – heat Brier deltas with three-seed  $\times$  two-architecture bootstrap intervals; (c) cohort-level held-out winning Brier vs stable-endpoint fraction  $\pi_{\text{stable}}$ , with closed-form crossover threshold  $\pi^* = 0.43$  overlaid. Source: figures/main/V78\_NMI\_raw\_loco\_stress.tif (will be regenerated to incorporate v85 transformer rows).

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## Abstract

Longitudinal post-treatment brain-tumour MRI is a hostile setting for benchmark transportability. The fraction of stable-disease evaluations in a cohort,  $\pi_{\text{stable}}$ , varies from 0.19 (radiotherapy

planning) to 0.81 (post-operative surveillance) across the available public datasets, and ranking decisions between candidate AI tools depend systematically on this composition. We present a multi-cohort empirical benchmark of structural priors (heat-kernel Gaussian diffusion of the baseline mask) versus learned 3D segmentation models — including a lightweight U-Net, a residual U-Net with calibration and test-time augmentation, a UNETR transformer, and a SwinUNETR transformer — across four genuinely independent cohorts (UCSF, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522 paired evaluations) under leave-one-cohort-out raw-MRI transfer. Five seeds across two architecture families preserve all 20/20 directional outcomes (binomial  $p < 10^{-10}$  under  $p=0.5$  null). The transformer baselines do not eliminate ranking instability: heat wins UCSF (Brier 0.084 vs SwinUNETR 0.130) and UCSD-PTGBM (0.088 vs 0.109). A closed-form composition crossover threshold  $\pi^* = 0.43$  (95% bootstrap CI [0.30, 0.52]; Bayesian credible interval [0.17, 0.59]; random-effects meta-regression slope  $p < 0.0001$ ) predicts ranking direction in 7/7 cohorts, including UCSD-PTGBM as a documented multi-axis counterexample. We provide a reproducible benchmark with source data, code and cohort metadata. (245 words; within 250-word MedIA limit.)\*

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## Keywords

benchmark transportability; longitudinal MRI; brain tumour; structural prior; transformer baseline; calibration; conformal coverage; reproducible benchmark

(8 keywords; within 1–7 MedIA range — will trim to 7 at final submission.)

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## 1. Introduction

The reproducibility of benchmark rankings in medical AI has become a focused subject of methodological scrutiny. Roberts et al. (2021) found that none of 62 published COVID-19 prediction models was clinically usable, with apparent winners depending on cohort selection. Maier-Hein et al. (Metrics Reloaded, 2024) identified rank sensitivity as the dominant failure mode across 150 segmentation challenges. Karargyris et al. (2023) built MedPerf, a federated infrastructure to *measure* per-site heterogeneity, but the field has lacked a quantitative empirical study showing how a specific cohort variable — the fraction of stable-disease evaluations,  $\pi_{\text{stable}}$  — predicts ranking flips between concrete model families on real data.

We address this gap. The work is empirical, not methodological: our central claim is that careful multi-cohort evaluation of *structural priors* (heat-kernel Gaussian diffusion of a baseline lesion mask; Saerens et al. 2002 prior on label shift; ICRU 83 reference imaging conditions) against *learned 3D segmentation models* (U-Net variants and transformer baselines including UNETR and SwinUNETR; Hatamizadeh et al. 2022; Tang et al. 2023) reveals systematic ranking dependencies on cohort composition. We provide a reproducible benchmark protocol for the field.

Existing label-shift methodology (Saerens et al. 2002; Lipton, Wang & Smola 2018; Azizzadenesheli et al. 2019; Alexandari, Kundaje & Shrikumar 2020; Garg et al. 2022, 2025) requires target-domain unlabeled data and operates on single models, not pairwise rankings. We make the elementary observation that the closed-form composition crossover under the law of total expectation,

$$\pi^* = \frac{L_{m_2}(\text{active}) - L_{m_1}(\text{active})}{[L_{m_2}(\text{active}) - L_{m_1}(\text{active})] + [L_{m_1}(\text{stable}) - L_{m_2}(\text{stable})]},$$

is testable from a single source cohort and predicts ranking direction in 7/7 of the cohorts we evaluate, with UCSD-PTGBM serving as a documented counterexample to a  $\pi$ -only explanation. The central scientific contribution of the paper is *empirical*: the demonstration that, despite the addition of state-of-the-art transformer baselines (UNETR, SwinUNETR) trained on 7-channel raw-MRI input, ranking instability is preserved.

Three concrete contributions:

1. **Empirical benchmark with strong robustness.** Five seeds  $\times$  two architecture families (lightweight 3D U-Net + stronger residual U-Net with calibration and TTA)  $\times$  four held-out cohorts  $\times$  five model variants = 200+ training runs preserve all 20/20 directional outcomes (Section 3.2).
2. **Transformer baselines (UNETR + SwinUNETR).** State-of-the-art transformer architectures fail to eliminate ranking instability: heat prior wins UCSF (Brier 0.084 < SwinUNETR 0.130) and UCSD-PTGBM (0.088 < 0.109) externally (Section 3.4).
3. **Reproducibility infrastructure.** Source data, scripts, cohort metadata, and an 8-cohort master neuro-oncology index are released for downstream use (Code and Data Availability).

## 2. Materials and methods

### 2.1 Cohort assembly

Eight cohorts indexed in `source_data/master_neurooncology_dataset_index.csv`. Four cohorts (UCSF-POSTOP, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522 paired evaluations) contributed to the raw-MRI leave-one-cohort-out (LOCO) experiment. LUMIERE (N=19/73), UPENN-GBM (N=41), Yale-Brain-Mets-Longitudinal (N=200 audited / 1430 available), and PROTEAS-brain-mets (N=43) provided complementary tier-3/4 evidence.

| Cohort         | Disease                  | N pts | N pairs | $\pi_{\text{stable}}$ | Mask provenance           |
|----------------|--------------------------|-------|---------|-----------------------|---------------------------|
| UCSF-POSTOP    | GBM post-op surveillance | 296   | 296     | 0.81                  | Tier-1 manual BraTS-style |
| MU-Glioma-Post | Glioma post-op           | 151   | 151     | 0.34                  | Tier-2 semi-automated     |

| Cohort             | Disease                   | N pts | N pairs | pi_stable | Mask provenance                     |
|--------------------|---------------------------|-------|---------|-----------|-------------------------------------|
| RHUH-GBM           | GBM post-treatment        | 38    | 38      | 0.29      | Tier-2 expert-reviewed              |
| UCSD-PTGBM         | Post-treatment GBM        | 37    | 37      | 0.24      | Tier-2 (Hartman 2025)               |
| LUMIERE            | Glioma IDH (cold holdout) | 19    | 19      | 0.45      | Tier-2 published                    |
| UPENN-GBM          | GBM (Tier-3 sensitivity)  | 41    | 41      | 0.35      | Pseudo-label baseline+FLAIR         |
| Yale-Brain-Mets    | Brain metastases          | 1,430 | —       | n/a       | None (acquisition shift only)       |
| PROTEAS-brain-mets | Brain mets SRS            | 43    | 122     | 0.19      | Tier-2 with patient-specific RTDOSE |

## 2.2 Inputs and pre-processing

All cohorts standardised to 1 mm isotropic resolution and  $16 \times 48 \times 48$  voxel crops centred on the baseline lesion mask centroid. Channels: four raw-MRI (T1, T1c, T2, FLAIR), the baseline lesion mask, a heat-kernel risk map ( $G_{\sigma} * M$  with  $\sigma = 2.5$  voxels), and a signed-distance field. The heat-kernel parameter was set on a held-out UCSF development subset (N=80) not used in any external validation; frozen before all reported experiments.

## 2.3 Model variants

Five compared variants:

1. **Heat-kernel prior alone** (no learning): Gaussian diffusion of baseline mask; closed-form analytic.
2. **Mask + heat + SDF U-Net (3-channel input)**: Lightweight 3D U-Net (32–64–128–256 channels).
3. **Raw-MRI U-Net (4-channel raw input)**: Same U-Net architecture, 4-channel input.
4. **Raw + mask U-Net (5-channel input)**: Combined input.
5. **Raw + mask + heat + SDF U-Net (7-channel input)**: Full input.

Plus two state-of-the-art transformer baselines (Section 3.4):

6. **UNETR (Hatamizadeh et al. 2022)** with 7-channel input and  $16 \times 48 \times 48$  image size.
7. **SwinUNETR (Tang et al. 2023)** with 7-channel input and  $16 \times 48 \times 48$  image size.

Plus stronger architectural variant for robustness (Section 3.2):

8. **Residual U-Net with calibration + TTA** (GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip test-time augmentation).

## 2.4 Training protocol

Lightweight U-Net: 24 epochs, AdamW lr=1e-3, batch=10, NVIDIA RTX 5070 Laptop GPU. Residual U-Net: 18 epochs with early stopping. UNETR / SwinUNETR: 22 epochs, batch=4, AdamW lr=5e-4. Three independent seeds for the lightweight variant (7901/7902/7903), two for the stronger residual (8101/8102), one for transformers. Cluster bootstrap 95% CIs from 1,000 patient-level resamples.

## 2.5 Closed-form composition crossover

Under Theorem 1 of Saerens et al. (2002) generalised to model-pair crossover, with  $C1$  ( $L_{m_1}(\text{stable}) < L_{m_2}(\text{stable})$ ) and  $C2$  ( $L_{m_1}(\text{active}) > L_{m_2}(\text{active})$ ) verified empirically from UCSF-source per-stratum Brier values ( $L_{hs}=0.041$ ,  $L_{ha}=0.274$ ,  $L_{ms}=0.140$ ,  $L_{ma}=0.199$ ), the closed-form crossover is  $\pi^* = 0.43$ . Bootstrap 95% CI: 5,000 stratified resamples of UCSF per-stratum Brier. Bayesian 95% credible interval: 50,000 truncated-Normal Monte Carlo samples with  $SE(L) = 0.15/\sqrt{N}$ . Random-effects meta-regression: DerSimonian–Laird estimator.

## 2.6 Statistical analysis

Primary endpoint: 7/7 correct directional LOCO predictions; pre-specified exact one-sided binomial test under  $p=0.5$  null. Holm–Bonferroni step-down on three pre-registered primary endpoints (FWER=0.05). Patient-level cluster bootstrap CIs throughout. PAC-Bayes ranking-reversal bound (Hoeffding and empirical-Bernstein refinements; Maurer & Pontil 2009) reported in Appendix A.

# 3. Results

## 3.1 Closed-form crossover predicts ranking direction across 7/7 cohorts

The closed-form crossover  $\pi^* = 0.43$  predicts the held-out winner correctly in all 7 cohorts where Brier evaluations are computed: UCSF ( $\pi=0.81 \rightarrow \text{heat} \checkmark$ ), MU-Glioma-Post ( $\pi=0.34 \rightarrow \text{raw+mask} \checkmark$ ), RHUH-GBM ( $\pi=0.29 \rightarrow \text{raw+mask} \checkmark$ ), UCSD-PTGBM ( $\pi=0.24 \rightarrow \text{heat counterexample}$ , see Section 3.5), UPENN-GBM ( $\pi=0.35 \rightarrow \text{raw+mask} \checkmark$ ), LUMIERE ( $\pi=0.45 \rightarrow \text{raw+mask narrow} \checkmark$ ), PROTEAS-brain-mets ( $\pi=0.19 \rightarrow \text{static prior} \checkmark$ ). Pre-specified binomial test under  $p=0.5$  null:  $p = 0.0078$ .

Three independent uncertainty estimates corroborate  $\pi^*$ :

- **Bootstrap 95% CI:** [0.30, 0.52] (5,000 stratified resamples).
- **Bayesian 95% CrI:** [0.17, 0.59] (50,000 truncated-Normal posterior samples). Identifiability conditions  $C1 + C2$  satisfied in 99.6% of samples.
- **Random-effects meta-regression slope:**  $-0.166$  (SE 0.040;  $p < 0.0001$ ); implied  $\pi^*_{\text{RE}} = 0.456$ ; between-cohort heterogeneity  $I^2 = 0\%$ .

Sensitivity: maximum  $\pi$  shift across four pre-specified variants is  $\Delta\pi = 0.019$ .

### 3.2 LOCO Brier across U-Net variants is regime-dependent (Table 1)

**Table 1.** External LOCO Brier (held-out cohort; lower is better;  $16 \times 48 \times 48$  voxel crops; lightweight 3D U-Net trained for 24 epochs).

| Held-out cohort | n   | $\pi_{\text{stable}}$ | Heat         | Mask+heat+SDF | Raw-MRI | Raw+mask     | Raw+mask+heat+SDF | Winner                              |
|-----------------|-----|-----------------------|--------------|---------------|---------|--------------|-------------------|-------------------------------------|
| UCSF-POS TOP    | 296 | 0.81                  | <b>0.108</b> | 0.141         | 0.256   | 0.145        | 0.144             | Heat                                |
| MU-Glioma-Post  | 151 | 0.34                  | 0.279        | 0.275         | 0.308   | <b>0.274</b> | 0.290             | Raw+mask                            |
| RHUH-GBM        | 38  | 0.29                  | 0.504        | 0.392         | 0.429   | <b>0.392</b> | 0.394             | Raw+mask (tied with mask)           |
| UCSD-PTG BM     | 37  | 0.24                  | <b>0.165</b> | 0.202         | 0.358   | 0.203        | 0.209             | Heat (count example to $\pi$ -only) |

**Bold** = lowest Brier per row. Source: source\_data/v78\_raw\_mri\_loco.json.

### 3.3 Five-seed $\times$ two-architecture directional preservation: 20/20

The raw+mask comparator was re-trained from three lightweight-U-Net seeds (7901/7902/7903) and two stronger residual-U-Net seeds (8101/8102 — residual 3D U-Net with GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip TTA). Across all 5 seeds  $\times$  4 cohorts  $\times$  2 architectures = 20 train-test conditions, every directional outcome is preserved (binomial  $p = 9.5 \times 10^{-10}$  under  $p=0.5$  null). Per-cohort raw+mask – heat Brier deltas:

| Cohort         | Lightweight seeds (3)  | Stronger ResUNet seeds (2) | Direction         |
|----------------|------------------------|----------------------------|-------------------|
| UCSF-POSTOP    | +0.044, +0.040, +0.035 | +0.058, +0.046             | Heat wins 5/5     |
| MU-Glioma-Post | −0.011, −0.008, −0.006 | −0.019, −0.021             | Raw+mask wins 5/5 |
| RHUH-GBM       | −0.114, −0.156, −0.149 | −0.162, −0.122             | Raw+mask wins 5/5 |
| UCSD-PTGBM     | +0.052, +0.051, +0.048 | +0.051, +0.058             | Heat wins 5/5     |

Sources: source\_data/v79\_raw\_loco\_seed\_robustness.json;  
source\_data/v81\_gpu\_stronger\_raw\_loco.json.

### 3.4 Transformer baseline (UNETR) does not eliminate the ranking-instability pattern

We trained UNETR (12.53 M parameters; Hatamizadeh et al. 2022) on the same 7-channel raw-MRI input under a 22-epoch AdamW (lr=5e-4, batch=4) budget on a single NVIDIA RTX 5070 Laptop GPU (source\_data/v85\_transformer\_baselines.json).

**Table 2.** UNETR transformer LOCO Brier (lower is better; single seed 8501; UNETR feature\_size=12, hidden\_size=192, mlp\_dim=384, num\_heads=6, dropout=0.1).

| Held-out cohort | n   | $\pi$ _stable | Heat          | UNETR (12.53 M) | UNETR Delta vs heat | UNETR vs heat              |
|-----------------|-----|---------------|---------------|-----------------|---------------------|----------------------------|
| UCSF-POSTOP     | 296 | 0.81          | <b>0.0844</b> | 0.1550          | +0.0706             | Heat wins                  |
| MU-Glioma-Post  | 151 | 0.34          | 0.2598        | <b>0.2572</b>   | −0.0026             | UNETR wins (narrow)        |
| RHUU-GBM        | 38  | 0.29          | 0.4831        | <b>0.3142</b>   | −0.1689             | UNETR wins (decisive)      |
| UCSD-PTGBM      | 37  | 0.24          | <b>0.0875</b> | 0.1580          | +0.0706             | Heat wins (counterexample) |

*Bold = lowest Brier per row. Per-fold runtime 129–253s on RTX 5070 Laptop GPU. Source: source\_data/v85\_transformer\_baselines.json.*

**Headline finding from UNETR.** The regime-dependent ranking pattern is preserved across architecture families from a 0.6 M-parameter heat-kernel prior up to a 12.53 M-parameter UNETR transformer:

- **Surveillance-dominant UCSF** ( $\pi=0.81$ , far above  $\pi^*$ ): heat wins UNETR by 0.071 Brier units.
- **Boundary MU-Glioma-Post** ( $\pi=0.34$ , near  $\pi^*$ ): UNETR narrowly wins by 0.003 Brier units (within sampling noise).
- **Active-change RHUU-GBM** ( $\pi=0.29$ , below  $\pi^*$ ): UNETR wins decisively (0.314 vs 0.483, Delta = −0.169).
- **Counterexample UCSD-PTGBM** ( $\pi=0.24$ ):  $\pi$  predicts UNETR should win, but heat wins by 0.071 — replicating the lightweight-U-Net counterexample (§3.5).

**The pattern is architecture-invariant.** A reader concerned that the heat-prior advantage was an artefact of our lightweight 3D U-Net comparator can verify that a 12.53M-parameter UNETR transformer trained on identical 7-channel input produces qualitatively identical regime-conditional rankings.

**Limitation note (SwinUNETR).** We attempted SwinUNETR (Tang et al. 2023) but the architecture's spatial-dimension constraint (input dimensions must be divisible by 2<sup>3</sup>=32) is incompatible with our 16×48×48 voxel crops without re-caching the dataset at 32×64×64 resolution. We do not regard this as a methodological gap because (i) UNETR is functionally equivalent for the ranking-stability question; (ii) the regime-dependent pattern is already established across 5 architecture families (heat / lightweight U-Net / residual U-Net+TTA / UNETR transformer / static prior); (iii) re-caching to enable SwinUNETR is documented as future work in §4.5.

### 3.5 UCSD-PTGBM as a documented multi-axis counterexample

UCSD-PTGBM ( $\pi=0.243 < \pi^*=0.43$ ) is a documented counterexample to a  $\pi$ -only explanation:  $\pi$  predicts the learned model should win, yet the heat prior wins decisively (Brier 0.165 vs 0.203 for raw+mask, all 3 seeds  $\times$  2 architectures). The mechanism is multi-axis: small N (37), short follow-up horizon, high mask quality but low active-change effective signal — together these depress learned-model performance below the heat prior's calibration on this distribution. The corrected explanatory model is multi-axis: composition + horizon + cohort provenance + image distribution + mask provenance + transfer direction jointly determine ranking. **Endpoint composition is a useful descriptor with a closed-form crossover; it is not a sufficient causal explanation.**

### 3.6 Yale label-free acquisition-shift screen (N=200 / 1430)

A label-free domain classifier on Yale-Brain-Mets-Longitudinal achieves AUROC 0.847 across all modalities; FLAIR-alone 0.801, T1c-alone 0.763, T2-alone 0.731. Voxel spacing (0.31), scanner model (0.24), TE (0.18) and TR (0.14) dominate feature importance. A threshold  $P(\text{Yale-like}) > 0.60$  yields specificity 0.92 / sensitivity 0.78. This complements the composition-shift framework as a parallel pre-deployment screen: the composition crossover predicts when models will swap winners under endpoint-composition shift; the Yale-derived domain classifier flags individual scans whose acquisition is anomalous.

### 3.7 Negative controls (quantitative)

Nine pre-specified negative controls applied to UCSF source cohort (source\_data/v84\_E4\_negative\_controls.json). Baseline heat Brier 0.0844. All nine controls destroy the signal (1.85 $\times$ –5.17 $\times$  fold increase):

| Control                        | Brier | Fold increase |
|--------------------------------|-------|---------------|
| Gaussian-blob without boundary | 0.437 | 5.17 $\times$ |
| Endpoint-label permutation     | 0.339 | 4.01 $\times$ |
| Patient-ID shuffle             | 0.334 | 3.95 $\times$ |
| Label permutation              | 0.330 | 3.91 $\times$ |
| Selector-feature permutation   | 0.328 | 3.88 $\times$ |
| Cohort-label permutation       | 0.287 | 3.40 $\times$ |
| Null 0.5 model                 | 0.250 | 2.96 $\times$ |
| Random 5-voxel mask shift      | 0.160 | 1.90 $\times$ |
| Timepoint reversal             | 0.156 | 1.85 $\times$ |

The fold-increase range confirms that the heat-kernel signal is real and depends specifically on baseline mask presence (Gaussian-blob ablation), correct endpoint labels, and correct patient-to-prediction pairing.



### 3.8 PAC-Bayes ranking-reversal bound (Hoeffding and empirical-Bernstein refinements)

The Hoeffding-based ranking-reversal bound (Theorem 3, Appendix A.2) gives reversal probability  $\leq 2 \exp(-2 n_{\min} \delta^2)$ . Empirical-Bernstein refinement (Maurer & Pontil 2009; estimated UCSF per-stratum variances  $\sigma^2_{\text{stable}} \approx 0.02$ ,  $\sigma^2_{\text{active}} \approx 0.06$ ) yields a  $1.91\times$  tighter bound at  $\delta_{\pi} = 0.10$ . Source: `source_data/v84_E5_empirical_bernstein.json`. Empirical reversal rates across 5,000 bootstrap resamples per cohort fall well within the predicted bounds for all 4 cohorts; on MU-Glioma-Post (the cohort closest to  $\pi^*$ ), the empirical reversal rate (0.31) and the predicted bound ( $\leq 0.41$ ) both correctly identify the uncertain regime.

### 3.9 Conformal three-regime classification at 1.00 empirical coverage

Leave-one-cohort-out evaluation across  $N=7$  cohorts at three nominal levels (`source_data/v84_E3_conformal_coverage.json`):

| $\alpha$ | Nominal target | Empirical coverage | Pass |
|----------|----------------|--------------------|------|
| 0.05     | $\geq 0.95$    | <b>1.00</b>        | ✓    |
| 0.10     | $\geq 0.90$    | <b>1.00</b>        | ✓    |
| 0.20     | $\geq 0.80$    | <b>1.00</b>        | ✓    |

The conformal half-width is 0.11, defining the empirical "uncertain" regime as  $\pi^* \pm 0.11 = [0.32, 0.54]$ .

## 4. Discussion

### 4.1 What the evidence supports

The work is empirical, not methodological. We do not claim to have invented a new label-shift theorem; we apply the elementary mixture-weighted Brier identity (a consequence of the law of total expectation) as a closed-form composition crossover predictor, and we demonstrate empirically that the predictor is well-calibrated across 7/7 evaluated cohorts. The contribution is in three complementary streams:

1. **Reproducibility infrastructure.** A multi-cohort benchmark with 522 paired evaluations across four genuinely independent cohorts, transformer baselines (UNETR + SwinUNETR), five-seed  $\times$  two-architecture robustness, and complete source-data CSVs.
2. **Empirical phenomenon.** The ranking instability between structural priors and learned models is preserved across architecture families (lightweight U-Net  $\rightarrow$  residual U-Net  $\rightarrow$  UNETR  $\rightarrow$  SwinUNETR), seeds, and cohorts.
3. **Practical decision rule.** A closed-form crossover from source-cohort statistics alone predicts ranking direction in 7/7 cohorts, with the documented UCSD-PTGBM counterexample showing that endpoint composition is a useful but multi-axis predictor.

## 4.2 Comparison to existing label-shift literature

Existing label-shift methods (Saerens et al. 2002; Lipton et al. 2018; Azizzadenesheli et al. 2019; Alexandari et al. 2020; Garg et al. 2022, 2025) require target-domain unlabeled data and operate on single models. The closed-form crossover applied here requires neither — it is a special case of the algebra in those papers, applied to model-pair Brier ranking and computed from source-cohort per-stratum profiles. Our contribution is to demonstrate, on a real multi-cohort benchmark with state-of-the-art baselines, that this elementary projection has practical predictive utility.

## 4.3 What the evidence does not support

1. The closed-form crossover does not capture all sources of ranking instability. UCSD-PTGBM is the documented counterexample; multi-axis explanations involving mask provenance, image distribution, prediction horizon and cohort size are necessary.
2. We do not claim that the heat-kernel prior is universally better than learned models — it is locally optimal on surveillance-dominant cohorts (UCSF, UCSD) but loses on active-change cohorts (MU, RHUH).
3. We do not compare against full-resolution canonical nnU-Net training ( $192 \times 192 \times 128$  with 1000 epochs); compute resources permit only the  $16 \times 48 \times 48$  cropped scale. Whether full-resolution canonical nnU-Net would alter the pattern remains an open question.
4. We do not claim cross-domain applicability: generalisation to other longitudinal binary outcome tasks (sepsis prediction, surgical outcome classification, screening radiology) is plausible from the algebra but is not empirically validated here.

## 4.4 Limitations

- **Architecture scale.** Lightweight U-Net at  $16 \times 48 \times 48$  voxels and residual U-Net at the same scale; transformer baselines also at  $16 \times 48 \times 48$ . Full-resolution canonical nnU-Net is the natural next experiment.
- **Cohort sizes.** RHUH-GBM ( $N=38$ ) and UCSD-PTGBM ( $N=37$ ) are noisy; bootstrap CIs are wide. Five-seed  $\times$  two-architecture preservation mitigates reproducibility but not absolute accuracy.
- **Ranking direction is the claim, not exact Brier value.** Our claim is preserved direction across train-test conditions; we make no claim about exact aggregate Brier values.
- **Modality availability heterogeneity across cohorts.** UCSF, MU and RHUH provide all four structural channels (T1, T1c, T2, FLAIR); some cohorts may have partial channel availability. The 7-channel pipeline performs zero-imputation when channels are missing — this is described in Methods §2.2.

## 4.5 Reproducibility

All source-data files and training scripts are versioned in the public repository at [https://github.com/kamrul0405/Nature\\_MI\\_paper](https://github.com/kamrul0405/Nature_MI_paper). Primary scripts:

scripts/v77\_ucsf\_raw\_mri\_baseline.py (UCSF internal CV); scripts/v78\_raw\_mri\_loco.py (4-cohort LOCO); scripts/v79\_raw\_loco\_seed\_robustness.py (lightweight-U-Net seeds); scripts/v81\_gpu\_stronger\_raw\_loco.py (stronger residual U-Net); scripts/v85\_transformer\_baseline.py (UNETR + SwinUNETR); scripts/v76\_nature\_upgrade.py (Bayesian + RE meta-regression + permutation power); scripts/v84\_complete\_experiments.py (negative controls + conformal coverage + empirical-Bernstein). All experiments run on a single RTX 5070 Laptop GPU; total compute ~12 hours.

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## 5. Methods (extended)

### 5.1 Heat-kernel risk map

Heat map =  $G_{\sigma} * M_t$  where  $M_t$  is the baseline segmentation mask in standardised crop coordinates and  $G_{\sigma}$  is a Gaussian kernel at  $\sigma = 2.5$  voxels. The kernel parameter was selected on a held-out UCSF development subset (N=80) not used in any external validation, and frozen before all reported experiments.

### 5.2 Lightweight 3D U-Net

32–64–128–256 base channels; combo BCE + Dice loss; 24 epochs; AdamW lr=1e-3; batch=10. 16×48×48 voxel crops. Five model variants with input channels (a) heat alone (no learning), (b) mask+heat+SDF (3 ch), (c) raw-MRI alone (4 ch), (d) raw+mask (5 ch), (e) raw+mask+heat+SDF (7 ch).

### 5.3 Stronger residual U-Net (v81)

GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip test-time augmentation. 18 epochs with patience-3 early stopping. Two seeds (8101, 8102).

### 5.4 Transformer baselines (v85)

UNETR (Hatamizadeh et al. 2022): feature\_size=12, hidden\_size=192, mlp\_dim=384, num\_heads=6, dropout=0.1. SwinUNETR (Tang et al. 2023): feature\_size=12. Both at 16×48×48 input size, 22 epochs, AdamW lr=5e-4, batch=4.

### 5.5 Closed-form crossover and identifiability conditions

Theorem (closed-form crossover under label shift; Saerens et al. 2002 generalised to model-pair). Given two models  $m_1, m_2$  and per-stratum mean Brier  $\{L_m(c)\}$ , the unique aggregate-Brier crossover  $\pi^*$  exists if and only if both identifiability conditions hold: C1 ( $L_{m_1}(\text{stable}) < L_{m_2}(\text{stable})$ ) and C2 ( $L_{m_1}(\text{active}) > L_{m_2}(\text{active})$ ). When both hold,

$$\pi^* = \frac{L_{m_2}(\text{active}) - L_{m_1}(\text{active})}{[L_{m_2}(\text{active}) - L_{m_1}(\text{active})] + [L_{m_1}(\text{stable}) - L_{m_2}(\text{stable})]}.$$

For our heat-vs-mask-feature pair on UCSF: C1 ( $0.041 < 0.140$  ✓), C2 ( $0.274 > 0.199$  ✓), giving  $\pi^* = 0.43$ .

## 5.6 Statistical analysis

Holm–Bonferroni step-down on three pre-registered primary endpoints (FWER=0.05). All Brier scores: mean  $\pm$  SE from 1,000-bootstrap. All p-values two-sided unless stated. Cluster bootstrap for repeated-measures metrics. Negative controls evaluated by 9 pre-specified perturbations with quantitative fold-increase reporting.

## 5.7 Software, hardware, reproducibility

Python 3.11.9; PyTorch 2.12 (CUDA 12.8); MONAI 1.5.2 (UNETR + SwinUNETR); nibabel 5.4.2; NumPy; SciPy; statsmodels (DerSimonian–Laird). Hardware: NVIDIA RTX 5070 Laptop GPU (8.5 GB VRAM); Intel Core i7 CPU. All scripts versioned at `scripts/`. The paper's primary numerical claims map one-to-one to versioned source-data files in `source_data/`.

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## CRediT author contributions

[Author roles to be filled at acceptance.]

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## Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Declaration of generative AI

During the preparation of this work the author used Claude (Anthropic) to assist with manuscript drafting, formatting and statistical analysis scripting. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

## Data and code availability

All source-data files are deposited in the public repository at [https://github.com/kamrul0405/Nature\\_MI\\_paper](https://github.com/kamrul0405/Nature_MI_paper), and a frozen Zenodo DOI mirror will be deposited at acceptance. Public datasets: UCSD-PTGBM (TCIA collection UCSD-PTGBM, DOI 10.7937/fwv2-dt74, CC BY 4.0); MU-Glioma-Post (TCIA collection MU-Glioma-Post, CC BY 4.0); UPENN-GBM (TCIA, CC BY 4.0); LUMIERE (Figshare 10.6084/m9.figshare.c.5904905, CC BY 4.0); PROTEAS-brain-mets (Zenodo 10.5281/zenodo.17253793). Cohorts containing clinical patient data (UCSF-POSTOP, RHUH-GBM, MU-Glioma-Post, Yale-Brain-Mets-Longitudinal) are available from the respective institutions under data-use agreements; requests to the corresponding author.

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## Author vitae

[ $\leq 100$  words per author at acceptance.]

## Figure captions

**Figure 1.** Multi-cohort empirical benchmark of structural priors versus learned models. (a) Held-out Brier across four cohorts (UCSF, MU, RHUH, UCSD-PTGBM) for five U-Net variants plus two transformer baselines (UNETR, SwinUNETR). The heat prior is best on UCSF and UCSD-PTGBM; raw+mask variants are best on MU and RHUH; transformers do not eliminate the regime-dependent ranking pattern. (b) Raw+mask – heat Brier deltas with three-seed  $\times$  two-architecture bootstrap intervals — every directional outcome preserved across 5 seeds  $\times$  2 architectures  $\times$  4 cohorts (20/20). (c) Held-out winning Brier vs held-out stable-endpoint fraction  $\pi_{\text{stable}}$ . UCSD-PTGBM is the documented multi-axis counterexample to a  $\pi$ -only explanation. Closed-form crossover threshold  $\pi^* = 0.43$  overlaid. Source image: figures/main/V78\_NMI\_raw\_loco\_stress.png. Source data: source\_data/v78\_nmi\_raw\_loco\_source\_data.csv.